BayesianEstimationTool Manual

BET fits models with population-wide parameters and individual-specific latent variables. This guide covers the actual model contracts, inference, array shapes, masking, and JAX modeling patterns.

1. Input and output contracts

`Estimator(data, ModelFn, localZNames=None, GlobalPriorFn=None, lkjConcentration=2.0, rngSeed=None)` takes a dictionary of arrays. The first axis of each data array indexes individuals; all arrays have the same number of individuals. Remaining axes can represent times, features, or spatial coordinates.

`ModelFn(globalParams, localZs, data)` receives dictionaries of global parameters, one individual's local latent values, and one individual's data. It returns a dictionary of model outputs. BET vectorizes this function across individuals and posterior draws; avoid Python side effects and conversions of traced JAX values to NumPy.

`GlobalPriorFn(dataShapes)` returns a dictionary of parameter names mapped to NumPyro distribution objects. Each shape in `dataShapes` excludes the individual axis. The global parameters are shared by the training population.

`TrainLikelihoodFn(globalParams, localZs, data, modelOut)` and `TestLikelihoodFn` each return a dictionary of site names mapped to `(observed_value, distribution)` tuples. A site may represent one scalar or a vector of observations; observation, distribution, and mask shapes must agree.

`ScoreFn(modelOut, data)` returns a scalar or array for one individual. BET vectorizes it over posterior samples and individuals.

2. Minimal working model

```
import jax.numpy as jnp
import numpyro.distributions as dist
from BayesianEstimationTool import Estimator, Svi

data = {"x": jnp.arange(8.0), "y": jnp.arange(8.0) + 0.2}

def global_prior(shapes):
```

```
    return {
        "slope": dist.Normal(1.0, 0.5),
        "offset_sd": dist.HalfNormal(0.5),
    }

def model_fn(global_params, local_zs, data):
    offset = global_params["offset_sd"] * local_zs["offset"]
    return {"prediction": global_params["slope"] * data["x"] + offset}

def likelihood(global_params, local_zs, data, model_out):
    return {"y": (data["y"], dist.Normal(model_out["prediction"], 0.5))}

def score(model_out, data):
    return jnp.abs(model_out["prediction"] - data["y"])

est = Estimator(data, model_fn, localZNames=["offset"],
                GlobalPriorFn=global_prior, rngSeed=1)
est.Train([0, 1, 2, 3, 4, 5], likelihood,
          Svi(numSteps=500, numSamples=100, progressBar=False))
train_scores = est.ScoreTrainSamples(score)
est.Test([6, 7], nLocalZSamples=100)
test_scores = est.ScoreTestSamples(score)
```

BET supplies one individual's data to `ModelFn`; do not index a patient axis within the function. This example defines a local offset because the current `Test()` implementation requires at least one name in `localZNames`. `test_scores` has global-state, local-draw, and test-entry axes before any score-specific axes; inspect these dimensions before summarizing.

3. Hierarchical local variables

`localZNames=["growth", "response"]` introduces standardized normal latent variables for each individual. With multiple local variables, BET uses an LKJ prior for their Cholesky correlation structure, controlled by `lkjConcentration`. The user model converts standardized values into meaningful parameters.

```
def model_fn(global_params, local_zs, data):
    growth = jnp.exp(global_params["log_growth"] + 0.3 * local_zs["growth"])
    return {"prediction": data["baseline"] * jnp.exp(growth * data["time"])}
```

BET does not automatically infer an independent population scale or mean for each local variable; define these transformations explicitly when required.

4. Training, testing, and score dimensions

`Train(trainDataIdxs, TrainLikelihoodFn, Method, printSummary=False, rngSeed=None, saveLocalZs=True, saveModelOut=False)` estimates parameters from the selected individuals using a `Nuts(...)` or `Svi(...)` method.

`Test(testDataIdxs, TestLikelihoodFn=None, Method=None, nGlobalSamples=None, nLocalZSamples=2000, printSummary=False, rngSeed=None, saveModelOut=False)` supports population prediction for new individuals when no test likelihood is supplied, or local-effect inference conditional on a test likelihood and Method. **The current implementation requires at least one local Z for `Test`.** `nGlobalSamples=None` uses posterior medians; a positive value samples stored training draws. `nLocalZSamples` applies to the no-likelihood mode.

`ScoreTrainSamples(ScoreFn, nSamples=None, rngSeed=None)` returns `[posterior_draw, train_individual, ...score_shape]`.

`ScoreTestSamples(ScoreFn, nSamples=None, rngSeed=None)` returns `[global_draw, local_draw, test_individual, ...score_shape]`. For local-free models, save model outputs if scoring a test run.

5. Variable-length sequences and masking

Pad individual time series to a common length `T`, storing `time`, `y`, and a boolean `valid` mask as `[N, T]` arrays. Padded entries are storage placeholders, not observations.

```
data = {
    "time": jnp.array([[0., 1., 2.], [0., 3., 0.]]),
    "y": jnp.array([[1.0, 1.2, 1.4], [0.9, 1.1, 0.0]]),
    "valid": jnp.array([[True, True, True], [True, True, False]]),
}
```

Use a masked NumPyro distribution and replace invalid observed values with finite, safe values before log-probability evaluation:

```
def likelihood(global_params, local_zs, data, model_out):
    valid = data["valid"]
    pred = model_out["prediction"]
    safe_y = jnp.where(valid, data["y"], pred)
    return {"y": (safe_y, dist.Normal(pred, 0.2).mask(valid))}
```

The model must still produce finite predictions at padded positions. A zero mask does not guarantee safety if an intermediate expression already contains `NaN` or infinity. Do not multiply a divergent log-probability by zero and assume its gradient is safe.

To hide additional data during testing, store a separate observation mask, such as `test_observed = valid & (time <= cutoff)`. The test likelihood uses this mask; scoring may use the original validity mask to evaluate held-out observations. Do not leak hidden measurements into the model inputs or preprocessing.

6. JAX differential equation solvers

The model function may call JAX-compatible ODE solvers, including `jax.experimental.ode.odeint` or Diffrax. Keep solver inputs JAX arrays, use differentiable operations, and return outputs at the observation times. Diffrax is an optional external dependency, not part of BET's installation requirements.

```
from jax.experimental.ode import odeint

def rhs(state, time, rate):
    return rate * state

def model_fn(global_params, local_zs, data):
    rate = jnp.exp(global_params["log_rate"] + local_zs["rate"])
    times = data["time"]  # sorted, finite, compatible with odeint
    trajectory = odeint(rhs, data["initial"], times, rate)
    return {"prediction": trajectory}
```

With padded time grids, ensure solver times satisfy the solver's ordering requirements. A common approach is to use a shared regular solver grid and interpolate to each individual's observation times. Beware duplicate or unsorted padded times.

7. Numerical stability and diagnostics

Scale observations and model outputs to reasonable ranges, and use likelihood scales reflecting measurement noise rather than arbitrary tiny numbers. Extremely large residuals or near-zero noise can produce unstable log-probabilities and gradients. Replace missing observations with safe values before masking, and ensure model outputs remain finite even at padded positions. Use `jax.debug.print`, `jax.numpy.isfinite`, and small synthetic examples to isolate shape or gradient errors.

For multiple individuals, check each axis and broadcasting rule explicitly. Test the model and likelihood on one individual before fitting the population. Use posterior predictive checks; a successful NUTS or SVI run does not guarantee a scientifically adequate model.

8. Persistence and custom methods

`est.Save("fit.npz", saveAllData=True)` saves inference state and optionally data. Restore with `Estimator.Load("fit.npz", ModelFn=model_fn, GlobalPriorFn=global_prior)` when data was saved, or provide `data=` when it was not. Supply the original model and prior functions again; they are not serialized as executable code.

Custom inference methods must implement `_Fit(ModelFn, rngKey, samplePrefixes)` and `_SaveArgs()`. Run `ValidateMethod(method)` before use. See `METHOD_AUTHORING.md` for the minimum viable implementation and optional features.

9. Test strategy

The regression suite exercises NUTS, SVI, scoring, validation, and Save/Load. Scientific validation additionally requires synthetic hierarchical parameter recovery, population-to-new-individual transfer with varying observation counts, and numerical equivalence after persistence. See `TESTING.md` for the test invocation.

10. Conceptual workflow and statistical interpretation

BET separates **model specification** from the recurring hierarchical inference workflow. The user supplies an entry-level deterministic model, an optional distributional prior for shared parameters, and an observation likelihood. BET handles population training, generation or conditioning of local effects for new entries, posterior sample bookkeeping, and scoring.

An entry is one independently indexed unit (for example, a patient, experiment, or simulated replicate). Arrays share a leading entry axis. Observations within an entry, such as repeated measurements over time, belong on trailing axes and are modeled jointly by that entry's likelihood. Treating repeated measurements as separate entries would change the hierarchical interpretation.

Population parameters versus local z-scores. Global parameters are shared across training entries. Local latent values represent entry-to-entry variation and have standardized normal marginal priors. For example, a model might define a positive rate as `rate_i =`

`exp(log_rate_mean + log_rate_sd * z_i)`. Here `z_i` is a z-score on the log-rate scale, while `rate_i` is lognormally distributed on its physical scale. The location and scale parameters are supplied by the user model or inferred through `GlobalPriorFn`; BET does not automatically invent a mean and scale for every local variable.

When there are multiple local variables, their joint distribution can be correlated through the learned Cholesky factor. A positive correlation means that entries with unusually high values of one standardized effect tend also to have high values of another, **before conditioning on their individual observations**. Posterior local values conditioned on data need not have sample mean zero or standard deviation one.

Training and new-entry inference are different tasks. Training uses selected entries' observations to estimate shared parameters, the correlation structure, and training-specific local effects. Testing uses that trained population state to generate predictions for previously unseen entries. Without a test likelihood, local effects are sampled from the learned population distribution. With a test likelihood and an inference Method, the local effects are conditioned on the new entry's available observations. A test set must not be included in training if the goal is out-of-sample evaluation.

11. Model design: step-by-step

1. Prepare `data` as a dictionary of equally long first-axis arrays. Decide which trailing axes represent time, features, or spatial positions.
2. Implement `ModelFn(globalParams, localZs, data)` for **one entry**. Return named arrays or scalars in a dictionary. Do not sample NumPyro random variables in this deterministic model function.
3. Implement `GlobalPriorFn(dataShapes)` when shared parameters must be inferred. Return NumPyro distribution objects keyed by unique parameter names. If there are no shared unknown parameters, omit the function.
4. Specify `localZNames` only for genuine entry-varying effects. Map those standardized effects to physical parameters inside `ModelFn`.
5. Implement the training likelihood as a dictionary of `(observation, distribution)` pairs. Use distributions with correct event and batch shapes; use explicit masks for absent observations.
6. Select training indices and a configured `Nuts(...)` or `Svi(...)` instance. Call `Train`, inspect diagnostics, and verify model behavior with posterior predictive checks.
7. Select held-out indices. Call `Test` without a likelihood for population prediction, or provide a test likelihood and Method for inference using partial observations.

8. Define a score function returning a scalar or array for one entry and evaluate `ScoreTrainSamples` or `ScoreTestSamples`. Summarize over posterior axes deliberately; do not confuse draw axes with entry axes.

A useful design rule is that **data determine the observation model, not the mechanistic model's parameter names**. Baseline covariates can enter `ModelFn` or global-parameter transformations, while observation masks determine which outcomes are used for inference.

12. Complete hierarchical example: varying intercepts

The following example illustrates a learned population center and spread, standardized individual offsets, training, and population prediction. It is intentionally small; for substantive analyses increase the SVI optimization steps and inspect diagnostics.

```
import jax.numpy as jnp
import numpyro.distributions as dist
from BayesianEstimationTool import Estimator, Svi

data = {
    "x": jnp.arange(12.0),
    "y": 1.0 + 0.4 * jnp.arange(12.0) + jnp.array([0.1, -0.1] * 6),
}

def global_prior(shapes):
    return {
        "intercept_mean": dist.Normal(0.0, 3.0),
        "intercept_sd": dist.HalfNormal(1.0),
        "slope": dist.Normal(0.0, 1.0),
    }

def model_fn(global_params, local_zs, data):
    intercept = global_params["intercept_mean"] + global_params["intercept_sd"]
* local_zs["intercept"]
    return {"prediction": intercept + global_params["slope"] * data["x"]}

def likelihood(global_params, local_zs, data, model_out):
    return {"y": (data["y"], dist.Normal(model_out["prediction"], 0.3))}

def score(model_out, data):
    return jnp.abs(model_out["prediction"] - data["y"])

est = Estimator(data, model_fn, localZNames=["intercept"],
GlobalPriorFn=global_prior, rngSeed=7)
```

```
est.Train(list(range(10)), likelihood, Svi(numSteps=500, numSamples=200,
progressBar=False))
est.Test([10, 11], nLocalZSamples=100)
train_error_samples = est.ScoreTrainSamples(score)
test_error_samples = est.ScoreTestSamples(score)
```

The final test call draws new local intercepts from the learned population; it does **not** use held-out `y` values to infer those intercepts. Consequently, population predictions can be much more uncertain than predictions after observing an individual's response. To condition on available observations, call `Test` with a test likelihood and a Method, and make sure that likelihood only exposes observations available at prediction time.

13. NUTS and SVI reference

`Nuts(numWarmup=2000, numSamples=2000, numChains=4, acceptProb=0.95, denseMass=False, medianSamples=50)` uses NumPyro's NUTS MCMC kernel with median-based initialization. `numWarmup` is the adaptation period; `numSamples` is the retained number of draws **per chain**; `numChains` controls chain count; `acceptProb` is the target acceptance probability; `denseMass` enables dense mass-matrix adaptation; and `medianSamples` controls the initialization strategy. Its reported summary includes effective sample size and R-hat. Inspect divergences, mixing, and posterior predictive adequacy rather than treating a successful run as validation.

`Svi(numSteps=10000, numSamples=2000, stepSize=0.01, numParticles=1, progressBar=True, medianSamples=50, multivariate=False)` optimizes a variational approximation with Adam and an ELBO objective. `numSteps` is the number of optimization updates; `numSamples` controls approximate posterior draws generated after optimization; `stepSize` is Adam's learning rate; `numParticles` controls Monte Carlo particles for the ELBO; and `medianSamples` controls initialization. `multivariate=False` uses `AutoNormal`; `True` uses `AutoMultivariateNormal`, which can represent posterior covariance but may be expensive in large hierarchical models.

Method selection. NUTS is often preferable when reliable posterior uncertainty is central and runtime is manageable. SVI can be faster for large problems, but an apparently stable loss curve does not establish calibrated uncertainty. Compare posterior predictive behavior and uncertainty against NUTS or simulated ground truth when feasible. BET supports configured Method instances; using Method classes can reconstruct previously saved configuration when available.

14. Detailed public API reference

The signatures below correspond to the public exports `Estimator`, `Nuts`, `Svi`, and `ValidateMethod` and the public Estimator methods.

Estimator construction

`Estimator(data, ModelFn, localZNames=None, GlobalPriorFn=None, lkjConcentration=2.0, rngSeed=None)`

- `data`: dictionary of arrays sharing a leading entry axis.
- `ModelFn`: deterministic per-entry model function returning a dictionary.
- `localZNames`: optional list of named standardized local effects.
- `GlobalPriorFn`: optional function returning named NumPyro distributions for shared parameters.
- `lkjConcentration`: concentration of the LKJ correlation prior when multiple local effects are used.
- `rngSeed`: optional reproducibility seed.

Train

`Train(trainDataIdxs, TrainLikelihoodFn, Method, printSummary=False, rngSeed=None, saveLocalZs=True, saveModelOut=False)`

- `trainDataIdxs`: selected training entries.
- `TrainLikelihoodFn`: per-entry likelihood factory.
- `Method`: inference Method instance or supported Method class.
- `printSummary`: request inference summary output.
- `rngSeed`: optional call-specific seed.
- `saveLocalZs`: retain local posterior draws for later use.
- `saveModelOut`: retain generated model outputs rather than recomputing them.

Test

`Test(testDataIdxs, TestLikelihoodFn=None, Method=None, nGlobalSamples=None, nLocalZSamples=2000, printSummary=False, rngSeed=None, saveModelOut=False)`

- `testDataIdxs`: entries for population prediction or local updating.

- `TestLikelihoodFn`: omitted for unconditioned population prediction; provided for observation-conditioned inference.
- `Method`: inference Method for observation-conditioned testing; may reuse saved configuration.
- `nGlobalSamples`: `None` uses posterior median population state; a positive integer selects trained posterior states.
- `nLocalZSamples`: number of local draws in no-likelihood testing.
- `printSummary`, `rngSeed`, `saveModelOut`: reporting, reproducibility, and output retention.

Scoring

`ScoreTrainSamples(ScoreFn, nSamples=None, rngSeed=None)` evaluates training posterior predictions, with a draw axis followed by training-entry axis and any score dimensions.

`ScoreTestSamples(ScoreFn, nSamples=None, rngSeed=None)` evaluates the most recent test state, with global-draw and local-draw axes followed by test-entry axis and any score dimensions. Do not collapse axes without checking which dimension represents which source of uncertainty.

Persistence

`Save(fileName, saveTrainLocalZs=True, saveTestLocalZs=True, saveTrainModelOut=False, saveTestModelOut=False, saveAllData=False)` writes a compressed state file, or returns a record if `fileName=None`. Saving data increases portability but also file size and potential data-sensitivity obligations. The save flags govern inclusion of local draws and model outputs.

`Estimator.Load(fileName, ModelFn, GlobalPriorFn=None, data=None)` reconstructs an estimator using explicitly supplied functions. Provide `data` unless it was included in the saved state. Passing `ModelFn=None` retrieves the saved record instead of reconstructing an executable estimator.

`ValidateMethod(method)` checks a custom Method's required interface; see `METHOD_AUTHORING.md` for implementation details. Public exports are defined in package `__init__.py`.

15. Common mistakes and troubleshooting

Shape mismatch. Print shapes before tracing, and start with a single entry. All top-level data arrays need the same entry count; per-entry model outputs must agree with likelihood observation

shapes. Broadcasting may silently produce a plausible but incorrect likelihood, so verify expected dimensions explicitly.

Local z-scores mistaken for physical parameters. A latent z-score has a standard-normal marginal prior, not the same units as a growth rate, drug sensitivity, or measurement scale. Make all transformations explicit and document their units.

Hidden test outcomes leaking into inference. Supplying a full `y` array in test data is not itself a leak if the test likelihood masks unavailable values and the mechanistic model does not read those outcomes. Check all preprocessing, features, and masks to ensure future measurements cannot affect prediction.

Missing data and NaNs. Masked observations still require finite values at points where JAX evaluates intermediate expressions. Use safe finite placeholders and keep ODE solver time grids valid even for padded observations.

Unexpectedly narrow intervals. Check whether `nGlobalSamples=None` has fixed population parameters at their posterior medians. That mode does not propagate global posterior uncertainty. Use positive `nGlobalSamples` when that uncertainty matters, and remember that approximate SVI posteriors may be underdispersed.

Poor convergence. Rescale observations, inspect priors and parameter identifiability, and check whether the mechanistic solver produces finite gradients. For NUTS, examine chain mixing and convergence diagnostics; for SVI, inspect the loss trajectory and sensitivity to initialization.

Save/Load mismatch. Restore the same model function, global prior function, data shapes, and compatible Method configuration. Python function objects are not serialized; names are metadata, not executable code.

16. Verification and further reading

The repository's `TESTING.md` explains how to run automated tests, while `METHOD_AUTHORING.md` documents the custom inference interface. The included `Examples/` directory contains NUTS, SVI, and Save/Load scripts. The one-page `docs/Cheatsheet.md` is a compact companion, not a replacement for this manual.

Automated smoke and integration tests demonstrate that supported workflows execute; they do not by themselves establish parameter identifiability, calibrated credible intervals, or validity of a particular scientific model. For research claims, add simulation-based recovery and posterior predictive checks suited to the application.

17. Return values, sample axes, and state

The estimator retains the most recent training and test state. A new call to `Train` clears the prior test state. A new call to `Test` replaces the previous test state. Score functions operate on this stored state; they do not refit the model.

- `Train(...)` returns the inference method's output object. For NUTS this is NumPyro's MCMC object; for SVI it is the SVI result containing optimized parameters and losses. BET also stores the posterior samples it needs for later testing and scoring.
- `Test(...)` without a likelihood performs population prediction and returns the model-output dictionary when `saveModelOut=True`; otherwise its result is `None`. With a test likelihood, it returns a list containing one inference output per selected global posterior draw. If `nGlobalSamples=None`, BET uses one median global state, so the list has one element.
- `ScoreTrainSamples(...)` returns an array with shape `[posterior_draw, training_entry, ...score_shape]`.
- `ScoreTestSamples(...)` returns an array with shape `[global_draw, local_draw, test_entry, ...score_shape]`. In the no-likelihood test mode, there is one global state if medians are used; with sampled global states, the leading axis indexes those states. The local-draw axis represents variation among newly generated or inferred local effects.
- `Save(path)` writes a compressed NPZ file and returns no meaningful value. `Save(None)` returns an in-memory nested record. `Estimator.Load(path, ModelFn=None)` returns the stored record as a dictionary; supplying a model function reconstructs an Estimator.

The precise model-output shape depends on what `ModelFn` returns. Always write down expected axes before aggregating: average over posterior draws to summarize uncertainty, and over entries only when that matches the scientific question. Avoid flattening global and local draws together without considering how they were generated.

18. Interpreting test-time uncertainty

BET can represent at least two distinct sources of predictive uncertainty: uncertainty in population-level parameters and uncertainty in a new entry's local effects. The current `Test` implementation requires at least one named local Z, even though `Estimator` and `Train` can be used without local Zs. Measurement noise and uncertainty in model structure add further uncertainty through the likelihood and scientific model.

With `nGlobalSamples=None`, Test uses the posterior median for global parameters and the learned Cholesky state rather than propagating variation across the training posterior. This is a

useful conditional prediction but can understate uncertainty in the population. With a positive `nGlobalSamples`, BET samples global/Cholesky states from the training posterior and performs test inference for each selected state. This can be computationally more expensive because the test inference is repeated for each selected global state.

When no test likelihood is supplied, local effects are drawn from the learned population distribution. When a likelihood is supplied, the test method updates local effects using the observations represented by that likelihood, conditional on the selected global state. If the scientific target is prediction after one or more measurements, ensure the likelihood contains only measurements available at that prediction time. Score held-out outcomes separately and do not use them for local inference.

A score array is not itself a posterior predictive interval. To report uncertainty for a prediction, retain posterior predictions or scores over draws and calculate summaries or quantiles along the correct draw axes. State explicitly whether population uncertainty was propagated and whether scores are absolute error, squared error, log score, or another metric.

19. Conditioning on partial observations: a concrete pattern

A common use case is to predict the rest of a trajectory after observing only its first few measurements. Use the test likelihood to expose only those measurements to inference. Keep the score function separate so that it can evaluate the entire trajectory, including held-out measurements.

```
def test_likelihood(global_params, local_zs, data, model_out):
    # Assumes each entry has observations on the same sorted time grid.
    n_observed = 3
    observed = jnp.arange(data["y"].shape[-1]) < n_observed
    prediction = model_out["prediction"]
    # Replace held-out values before likelihood evaluation. The mask controls
    # which finite values contribute to the log likelihood.
    safe_y = jnp.where(observed, data["y"], prediction)
    return {
        "y": (safe_y, dist.Normal(prediction, 0.2).mask(observed))
    }

est.Test(testDataIdxs, TestLikelihoodFn=test_likelihood,
         Method=Svi(numSteps=1000, numSamples=200, progressBar=False))
heldout_scores = est.ScoreTestSamples(score)
```

Here `testDataIdxs` must identify entries excluded from training. The example assumes `data["y"]` is a time series and `model_out["prediction"]` has the same shape. Adapt the mask to each entry's actual observation schedule when lengths differ. If observations are irregular or have different valid times, use a validity mask combined with the observation cutoff. Ensure the deterministic model does not use future outcomes as inputs, and make every intermediate calculation finite at padded positions.

This call conditions local effects on the first three measurements, conditional on the selected global parameter state. With `nGlobalSamples=None` (the default), BET uses posterior medians for global parameters and the Cholesky factor; it does not integrate over their posterior variation. Set `nGlobalSamples` to a positive integer to repeat local inference across sampled global posterior states. This can be expensive, because BET runs the test inference separately for each selected state.